

# Mohammad Choupan Results:

## Model: SAG-POOL

### Model Net10 :

Train: 93%

Validation : 7% ~260

Test: Test Folder

Random seed : 42 - numpy,Torch

Epochs: 200

### Model Net40:

Train: 97%

Validation : 3%

Test: Test Folder

Random seed : 42 - numpy,Torch

Epochs: 200

### Tuning :

**Features: (x,y,z)**

| Dataset    | Learning rate | Weight decay | Accuracy% |
|------------|---------------|--------------|-----------|
| ModelNet10 | 0.01          | 0.0005       | 94.27     |
| ModelNet10 | 0.02          | 0.0005       | 94.27     |
| ModelNet10 | 0.03          | 0.0005       | 93.55     |
| ModelNet10 | 0.01          | 0.0003       | 94.27     |
| ModelNet10 | 0.01          | 0.0008       | 93.55     |
| ModelNet10 | 0.025         | 0.0005       | 93.55     |
| ModelNet10 | 0.01          | 0.0002       | 91.76     |
| ModelNet40 | 0.02          | 0.0005       | 79.54     |
| ModelNet40 | 0.005         | 0.0005       | 82.87     |
| ModelNet40 | 0.01          | 0.0005       | 82.15     |
| ModelNet40 | 0.01          | 0.0003       | 82.15     |
| ModelNet40 | 0.01          | 0.0008       | 79.83     |
| ModelNet40 | 0.005         | 0.0003       | 79.83     |
| ModelNet40 | 0.004         | 0.0005       | 83.14     |

**Final Result:**

| Dataset    | Feature      | Learning rate | Weight decay | Accuracy% |
|------------|--------------|---------------|--------------|-----------|
| ModelNet10 | x,y,z        | 0.01          | 0.005        | 87.11     |
| ModelNet10 | Between      | 0.01          | 0.005        | 86.12     |
| ModelNet10 | Closeness    | 0.01          | 0.005        | 83.26     |
| ModelNet10 | Eigen Vector | 0.01          | 0.005        | 87.33     |
| ModelNet10 | Harmonic     | 0.01          | 0.005        | 87.89     |
| ModelNet10 | Katz         | 0.01          | 0.005        | 83.04     |
| ModelNet10 | Load         | 0.01          | 0.005        | 87.56     |
| ModelNet10 | Page Rank    | 0.01          | 0.005        | 87.22     |
| ModelNet10 | all          | 0.01          | 0.005        | 87.78     |
| ModelNet40 | x,y,z        | 0.004         | 0.005        | 78.44     |
| ModelNet40 | Between      | 0.004         | 0.005        | 74.11     |
| ModelNet40 | Closeness    | 0.004         | 0.005        | 74.8      |
| ModelNet40 | Eigen Vector | 0.004         | 0.005        | 68.48     |
| ModelNet40 | Harmonic     | 0.004         | 0.005        | 77.03     |
| ModelNet40 | Katz         | 0.004         | 0.005        | 71.47     |
| ModelNet40 | Load         | 0.004         | 0.005        | 71.8      |
| ModelNet40 | Page Rank    | 0.004         | 0.005        | 71.27     |
| ModelNet40 | all          | 0.004         | 0.005        | 75.04     |

## Mohadese Atyabi Results:

**Model : Point-Net**

**Tuning:**

| Dataset    | Learning rate | Weight decay | Accuracy |
|------------|---------------|--------------|----------|
| ModelNet10 | 0.1           | 0.00001      | 92.18    |
| ModelNet10 | 0.2           | 0.00001      | 91.19    |
| ModelNet10 | 0.01          | 0.00001      | 92.18    |
| ModelNet10 | 0.02          | 0.0001       | 92.62    |
| ModelNet10 | 0.02          | 0.001        | 91.85    |
| ModelNet10 | 0.02          | 0.00001      | 92.95    |
| ModelNet40 | 0.1           | 0.00001      | 83.71    |
| ModelNet40 | 0.2           | 0.00001      | 77.31    |
| ModelNet40 | 0.02          | 0.0001       | 87.44    |
| ModelNet40 | 0.02          | 0.001        | 86.67    |
| ModelNet40 | 0.02          | 0.00001      | 86.35    |

## Final Result:

| Dataset    | Feature                | Learning rate | Weight decay | Accuracy |
|------------|------------------------|---------------|--------------|----------|
| ModelNet10 | Betweenness            | 0.02          | 0.00001      | 92.29    |
| ModelNet10 | Closeness centrality   | 0.02          | 0.00001      | 92.73    |
| ModelNet10 | Eigenvector centrality | 0.02          | 0.00001      | 92.51    |
| ModelNet10 | Harmonic centrality    | 0.02          | 0.00001      | 92.29    |
| ModelNet10 | Katz centrality        | 0.02          | 0.00001      | 92.62    |
| ModelNet10 | Load centrality        | 0.02          | 0.00001      | 92.29    |
| ModelNet10 | Pagerank               | 0.02          | 0.00001      | 92.4     |

